Assignment 1

Date Fady mahrous samir 230103236

Ex1:

Range = Highest − Lowest = 7 − 0 = 7

classes = 8

classwidth = $\frac{R}{C} = \frac{7}{8} = 0,875 \simeq 1$

class limits	Tally	Frequency
0	||	2
1	卌	5
2	卌 卌卌 卌||||	24
3	卌 |||	8
4	卌 |	6
5	||||	4
6		0
7	|	1
		50

the highest frequency is 22 which occurs $\boxed{\text{at 2 days per week}}$

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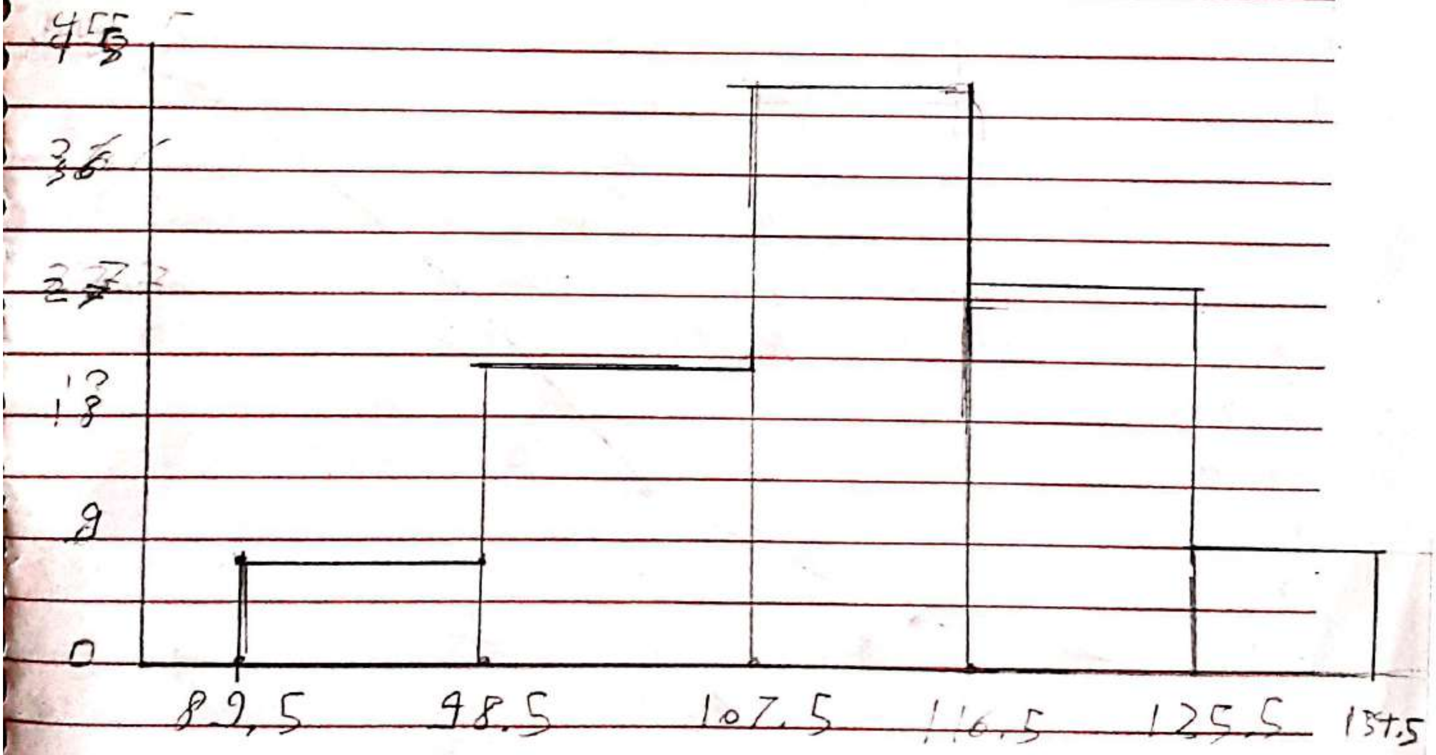
Ex2:

1. define class boundaries, midpoints, cumulative freq

class boundaries	midpoint	cumulative freq
89.5 _ 98.5	$\frac{H+L}{2} = 94$	0
98.5 - 107.5	103	6
107.5 - 116.5	112	28
116.5 - 125.5	121	77
125.5 - 134.5	130	99 - 108

2. Histogram



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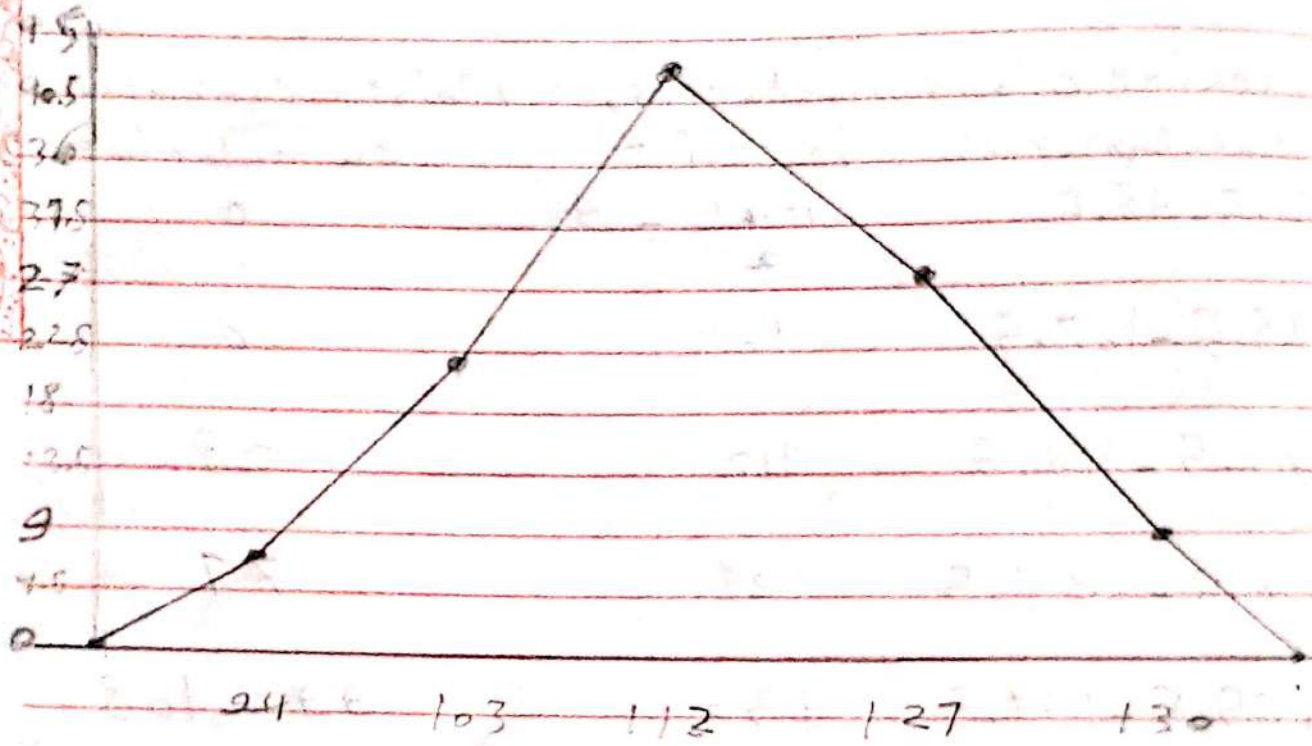
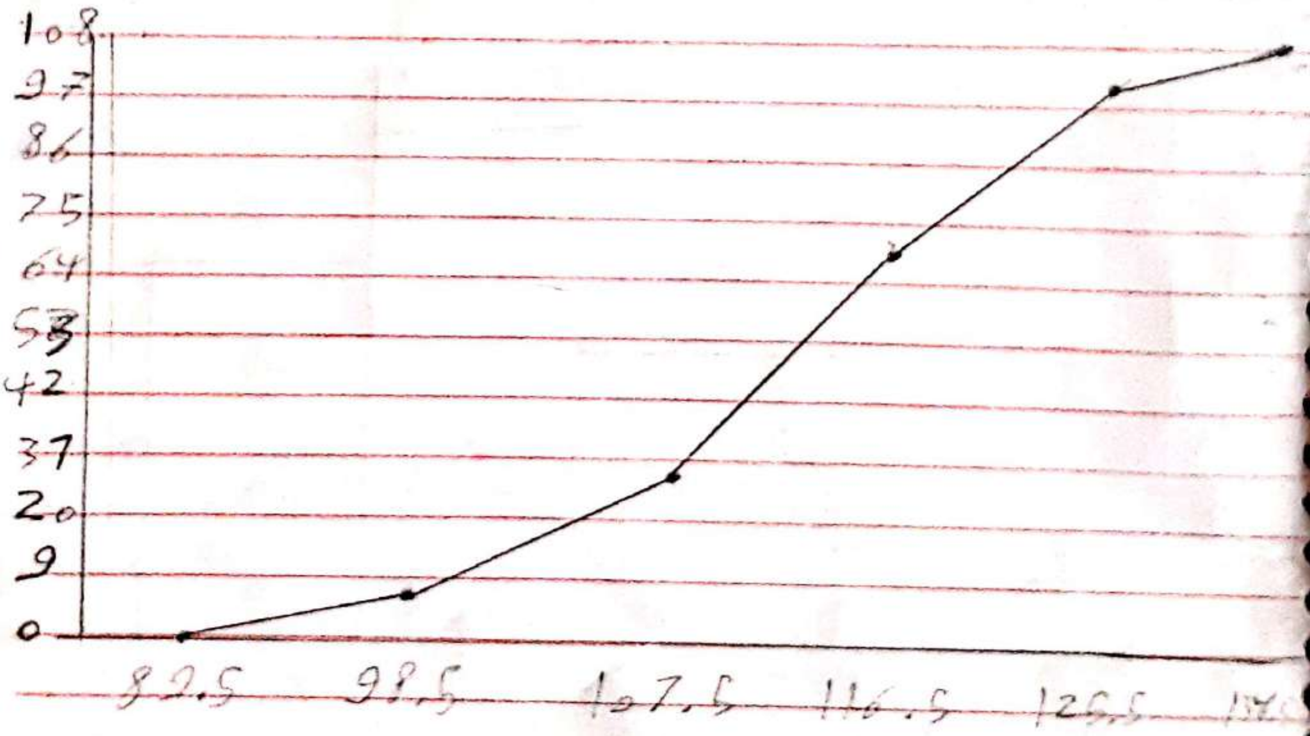
3-Frequency Polygon



4. ogive



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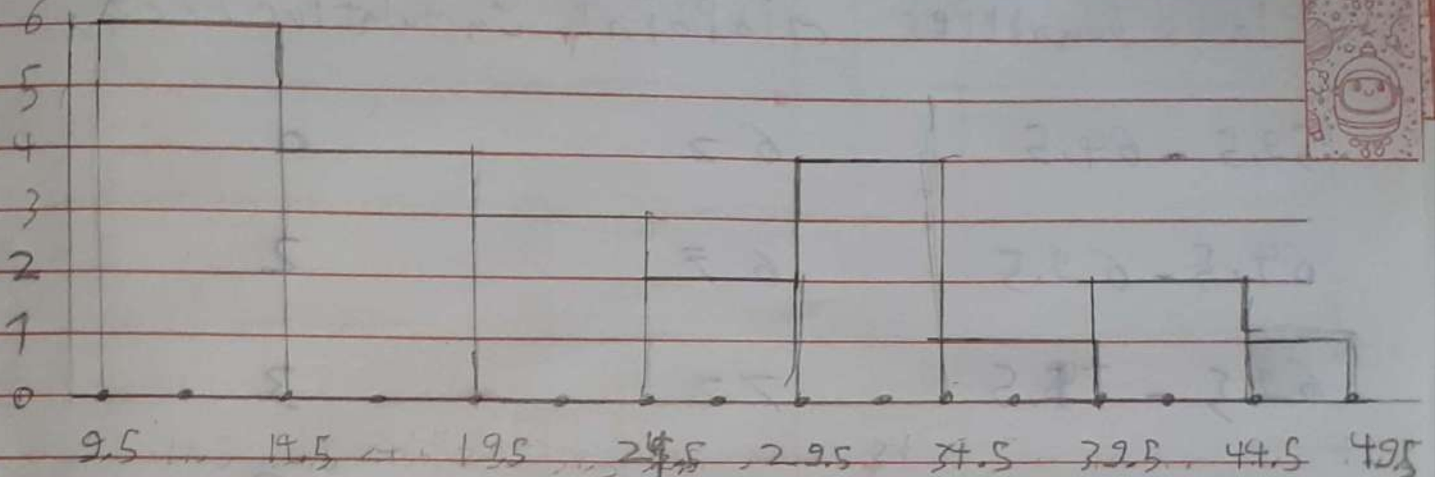
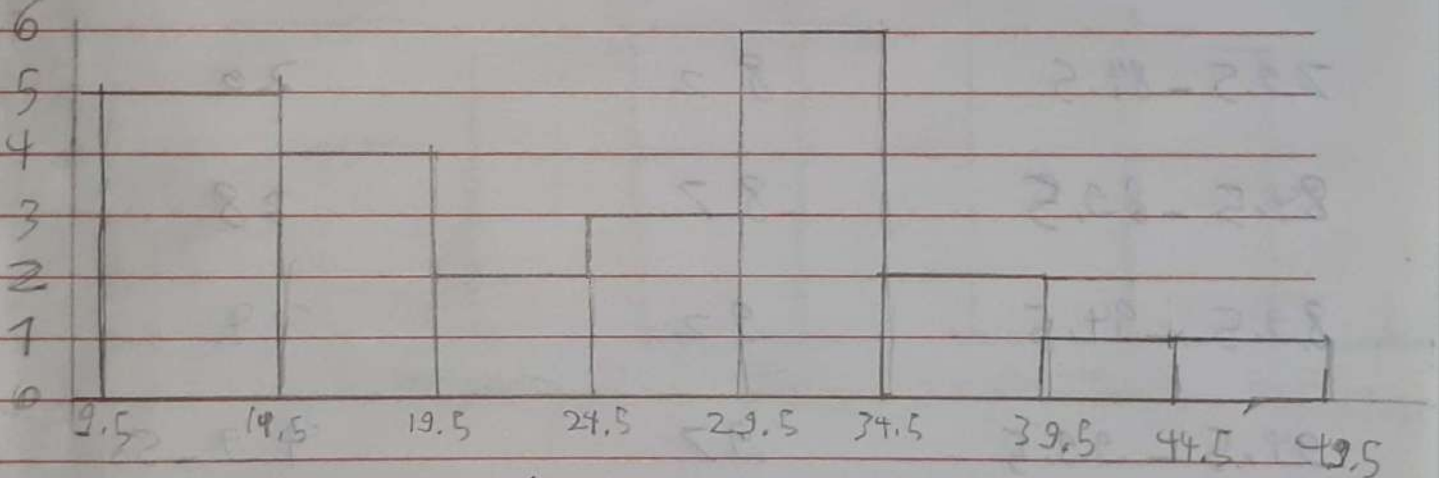
Ex3:

1, now, define class boundries



2- 5 years ago



the Pollution has slightly decreased over the past 5 years.

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Ex4:

1- define class ~~limits~~ boundries, midpoints, cumulative freq

class boundries	midpoint	cumulative freq
59.5 - 64.5	62	0
64.5 - 69.5	67	2
69.5 - 74.5	72	3
74.5 - 79.5	77	8
79.5 - 84.5	82	20
84.5 - 89.5	87	38
89.5 - 94.5	92	44
94.5 - 99.5	97	49 - 50

$\boxed{\text{most patients fall into between 80 \& 84}}$

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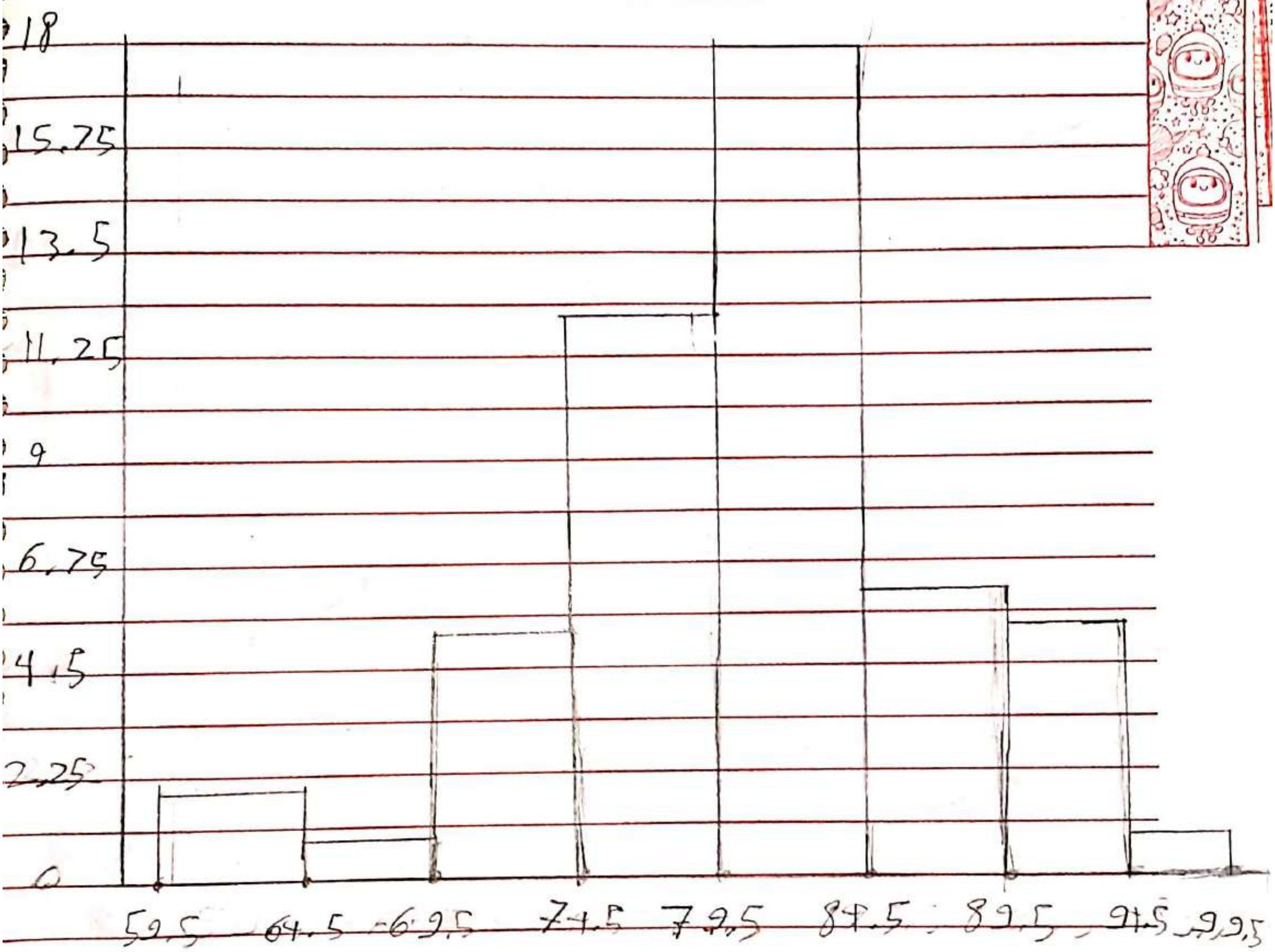
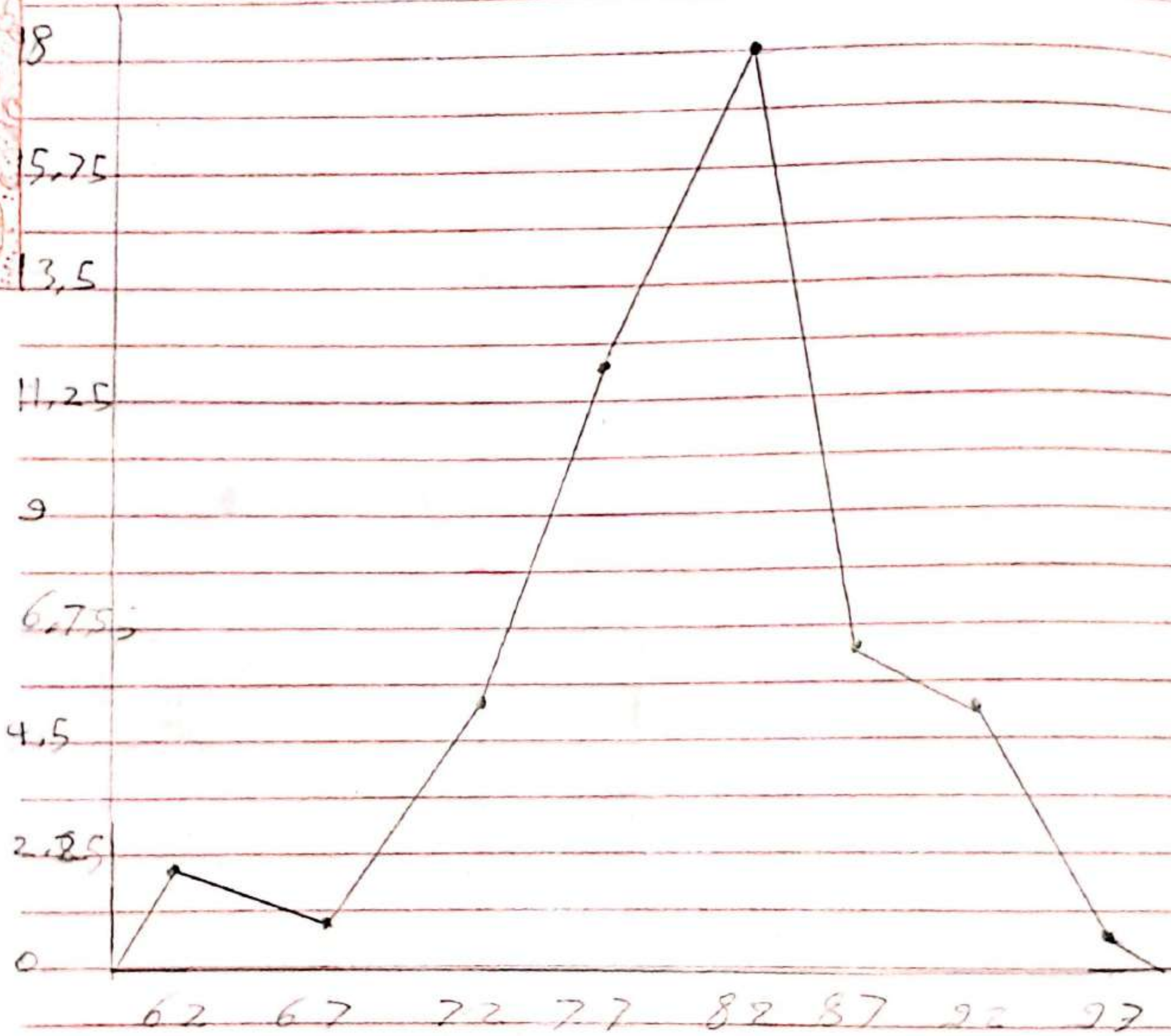
2. Histogram



3. Polygon



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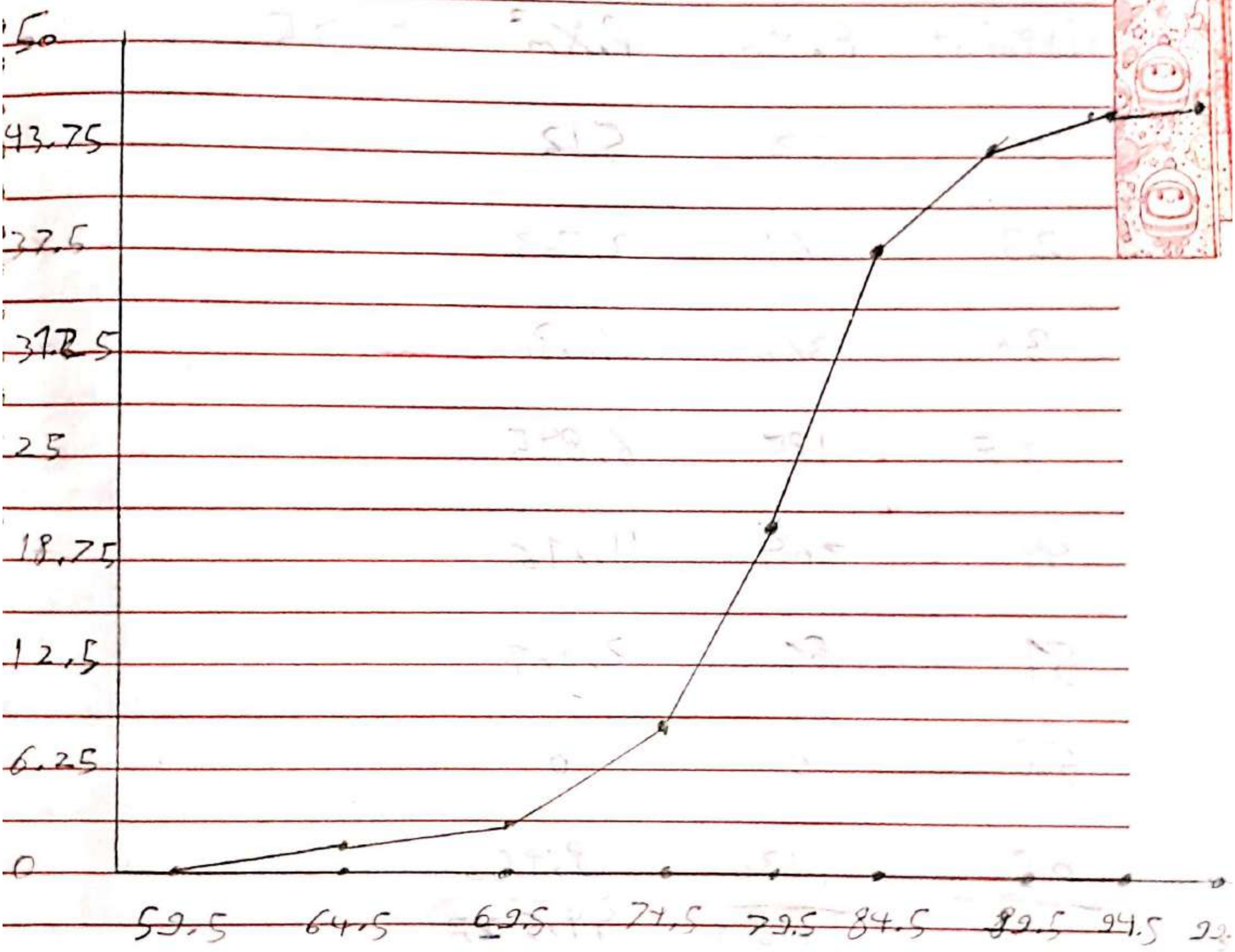
4.ogive



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Ex 5:

a.

Class limits	Frequency	midpoint	cumulative freq
22 – 24	1	23	0
25 – 27	3	26	1
28 – 30	0	29	4
31 – 33	6	32	4
34 – 36	5	35	10
37 – 39	3	38	15
40 – 42	2	41	18 – 20
	20		

~~Polygon~~

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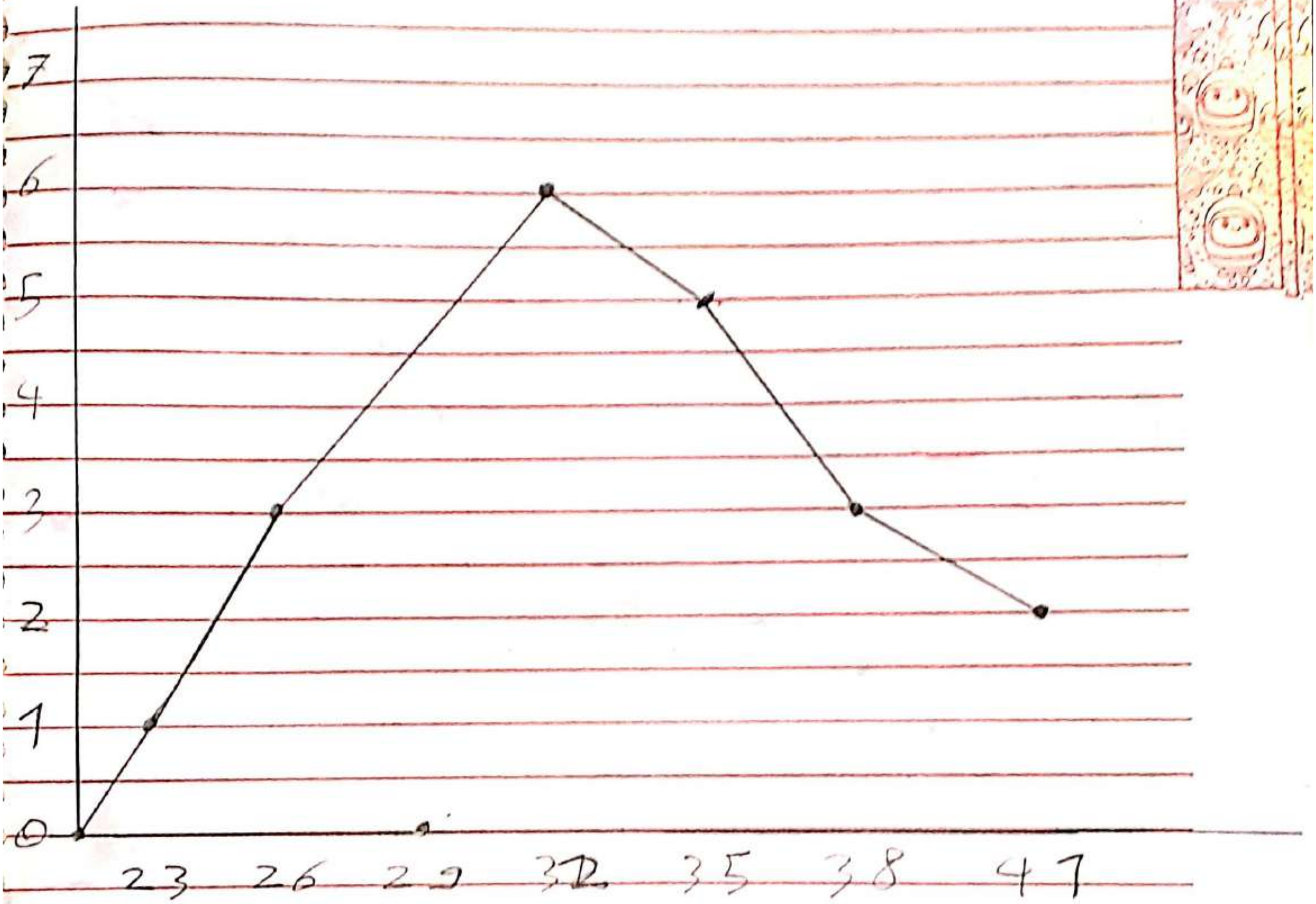
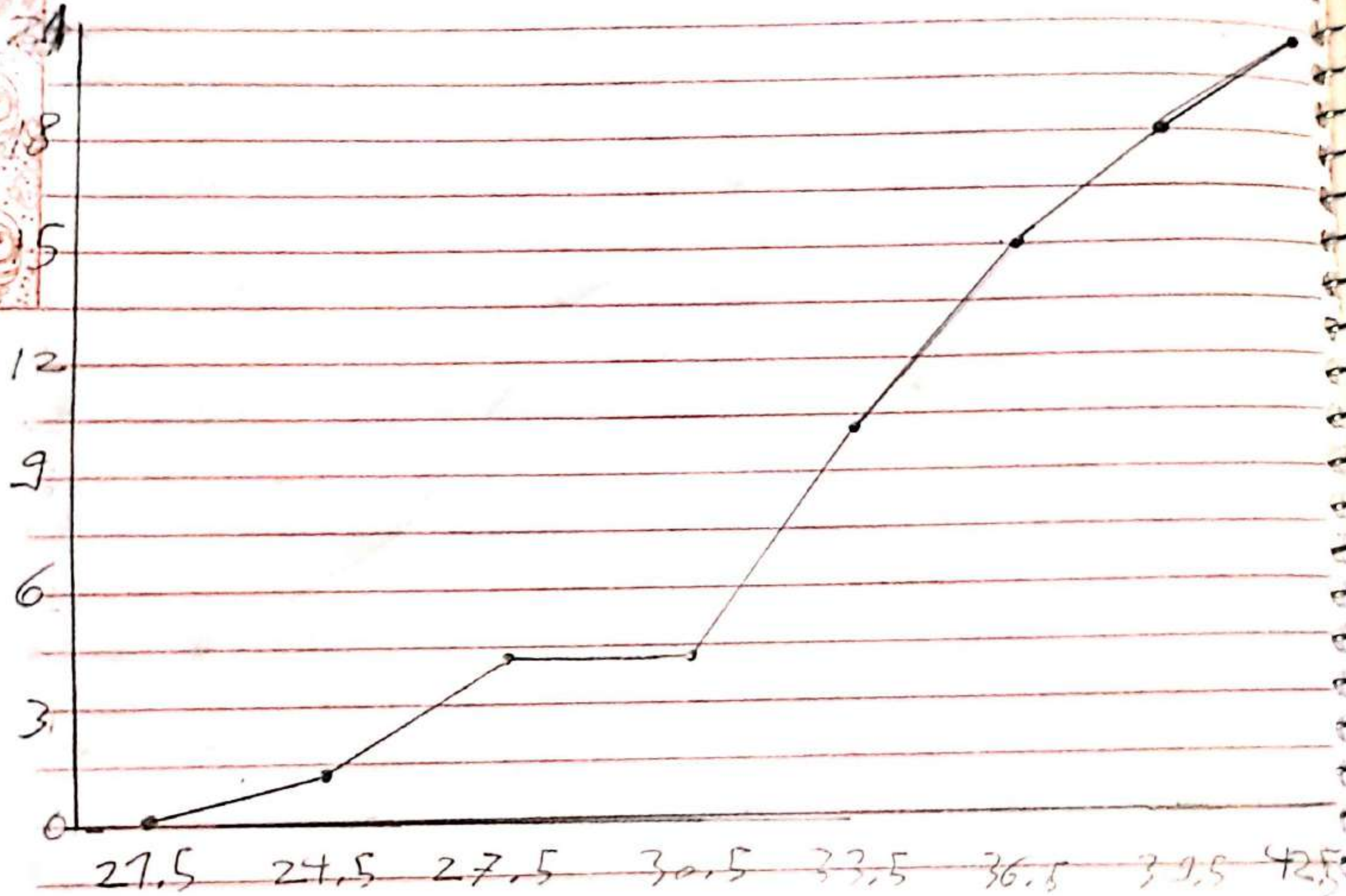
6. Polygon



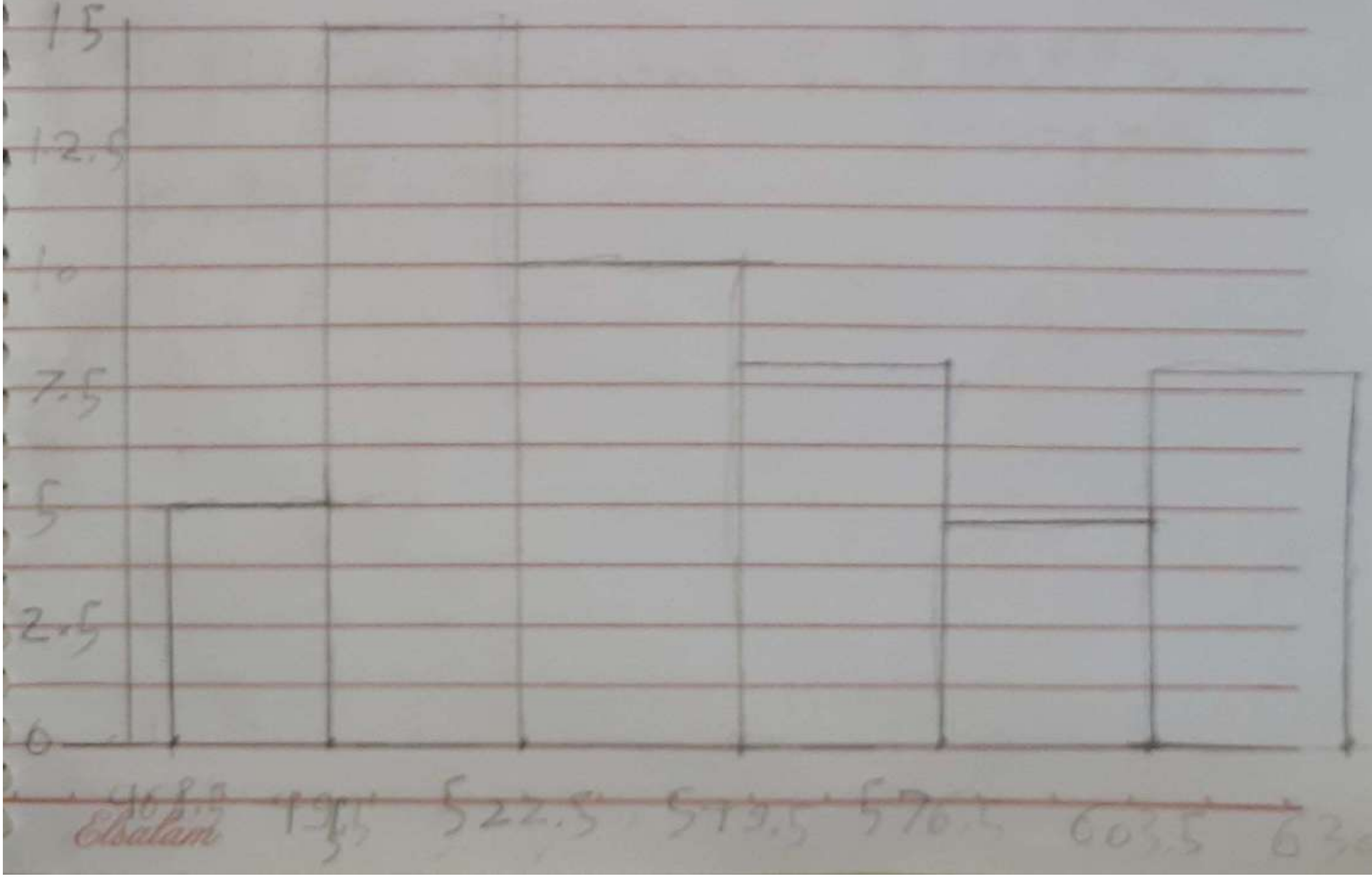
Ogive



Ex6:

class limits	frequency	midpoint
469 — 495	5	482
496 — 522	15	509
523 — 549	10	536
550 — 576	8	563
577 — 603	4	590
604 — 630	8	617
	50	

1. histogram



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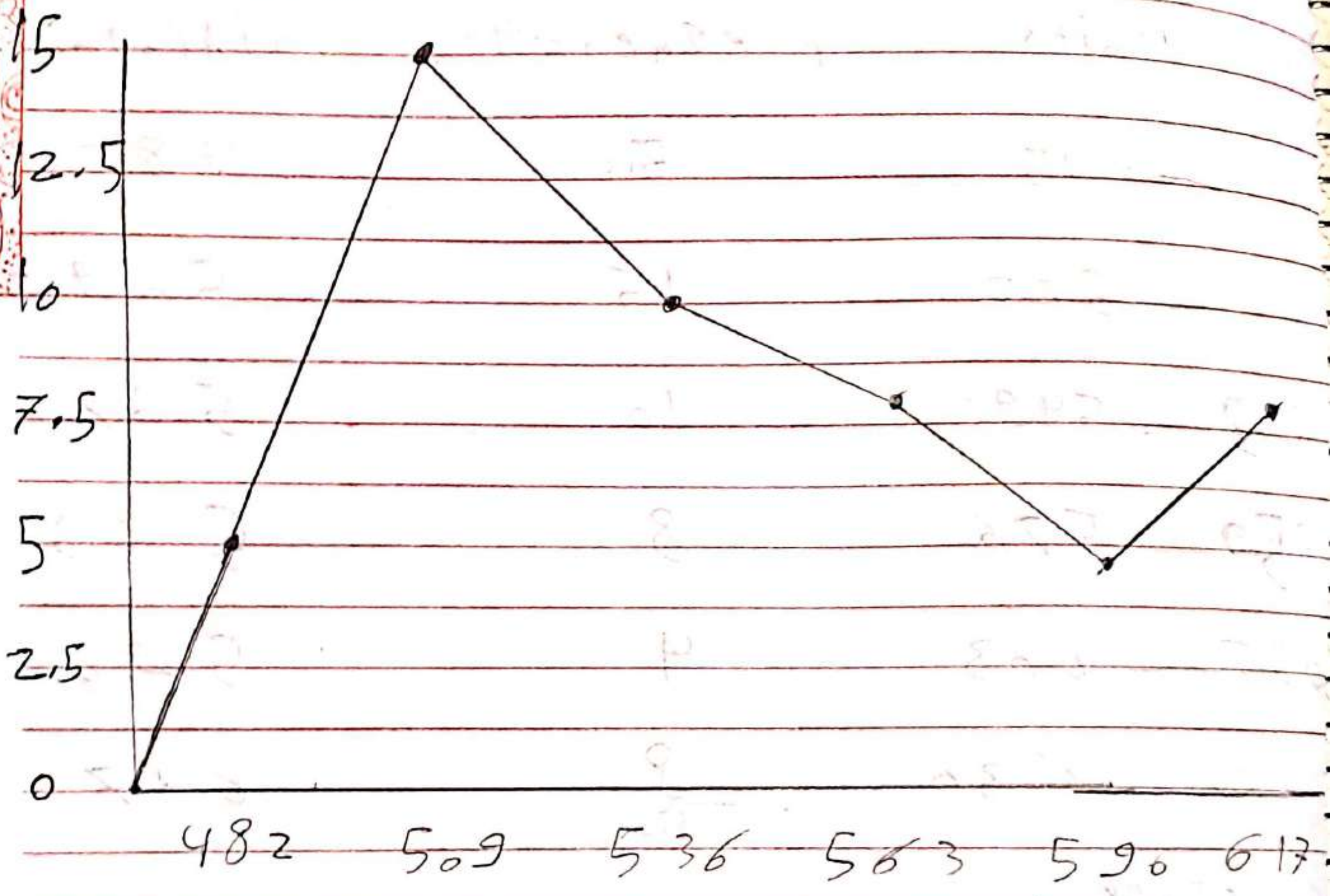
2. Polygon



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Ex7:

1. define midpoint, (f × x_m) → freq × midpoint

midpoint	(f × X_m)	
10	30	$\bar{X} = \frac{\sum(f \times X_m)}{\sum f} = \frac{590}{30}$
		$= 19.67$
15	75	
		modal class = 17.5 – 22.5
20	300	
25	125	
30	60	
	590	

Ex8.

midpoint	(f × X_m)	
10	120	$\bar{X} = \frac{\sum(f \times X_m)}{\sum f} = \frac{1022}{30}$
29	203	$= 34.07$
48	240	
		modal class = 0.5 – 19.5
67	201	
86	258	
	1022	

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Ex9:

midPoint	$f \times X_m$
17	238
20	240
23	414
26	260
29	435
32	192
	1779

$\bar{X} = \frac{\sum (f \times X_m)}{\sum f} = \frac{1779}{75}$

$= 23.72$

modal class $= 21.5 - 24.5$

Ex10:

$\bar{x} = 30$

a. $\bar{x} = \frac{20+30+40+50+60}{5} = 40$

b. $\bar{x} = 20$

c. $\bar{x} = \frac{100+200+300+400+500}{5} = 300$

d. $\bar{x} = \frac{1+2+3+4+5}{5} = 3$

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e. when a constant is added, subtracted, multiplied or divided by each value, the mean is affected in the same way by that constant

Ex 11:

1. men data set

a. range = 71

b. variance $= \sigma^2 = \frac{\sum (x-\mu)^2}{N} = \frac{4021.6}{10} = 402.16$

$\mu = \frac{823+837+858+797+812+814+832+862+833+826}{10}$

= 828.8

$(x-\mu)^2$	c. standard Deviation $= \sigma = \sqrt{402.16}$
33.64	$= 20.05$
67.24	
852.64	
1428.84	
282.24	
219.04	
10.24	
1102.24	
17.64	
7.84	
4021.6	

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2. women data set

a. range = 80

b. variance = $\sigma^2 = \frac{5542.7}{10} = 554.27$

$\mu = 769.7$

$(x-\mu)^2$
313.29
32.49
246.49
18.49
9.69
610.09
1135.69
2143.69
497.29
542.89
5542.7

c. Standard Deviation = σ = 23.54

the women's data set more variable

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Ex12:

midpoint	$f.X_m$	$f.X_m^3$
16	32	512
23	161	3,703
30	360	10,800
37	185	6,845
44	264	11,616
51	51	2,601
58	0	0
65	130	8,450
	1183	44,527

$n=35$

$$s^2 = \frac{35(44,527)-(1183)^2}{35(34)} = 133.6$$

$$s = \sqrt{133.6} = 11.56$$

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Ex13:

1. Construction workers

$C_{var} = \frac{S}{\bar{x}} \times 100 = \frac{2.3}{11} \times 100 = 20.9\%$

2. Factory workers

$C_{var} = \frac{1.8}{8} \times 100 = 22.5\%$

the factory workers are more variable

Ex14:

Admitted Patients:

$C_{var} = \frac{10.5}{80.2} \times 100 = 13.1\%$

2. Discharged Patients:

$C_{var} = \frac{18.3}{120.6} \times 100 = 15.2\%$

the Discharged Patients are more variable

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Ex15:

~~mean=3, s=32, k=~~

mean=180 min, s=32 min, k=3

$1-\frac{1}{k^2} = 88.89\%$, $-\frac{1}{k^2} = 0.8889 - 1 = -k^2 = -0.1111$

$k^2 = \frac{1}{0.1111} = 9$, $k=3$

chebyshev's form $= \bar{x} \pm k(s)$

$180 + 3\times32 = ~~84 \text{ min}~~ 276$ min

and

$180 - 3\times32 = 84$ min

$\boxed{\text{Range} = 84 — 276 \text{ min}}$

Ex16:

mean = 514, s = 40, k = 1; empirical rule

1. boundaries between 514-40 & 514+40 = 474 — 554

2. $514 \pm k(40) = 594$, $k=2$, Percentage = 95%, empirical rule

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Ex17:

a. Empirical Probability

b. Classical Probability

c. Empirical Probility

d. Classical Probability

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Ex18:

$P(E) = \frac{n(E)}{n(S)}$

a. $\frac{573079 + 775424}{573079 + 775424 + 211381 + 301264 + 24347 + 21683}$

$= 0.707 = 70.7\%$

b. $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$= \frac{46024 + 1098371 - 21683}{1907172} = 0.589 = 58.9\%$

c. $P = \frac{21683}{1907172} = 0.0114 = 1.1\%$

d. $P = 1 - P(\text{not At master}) = 1 - \frac{512645}{1907172}$

$= 0.731 = 73.1\%$

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Ex 19:

a. $\frac{2}{30} = 0.0667 = 6.7\%$

b. $\frac{10}{30} = 0.333 = 33.3\%$

c. $\frac{25}{30} = 0.833 = 83.3\%$

d. $\frac{25}{30} = 0.833 = 83.3\%$

e. $\frac{8+2}{30} = 0.333 = 33.3\%$

Ex 20:

a. independent

b. dependent

c. dependent

d. dependent

Ex 21:

35%

$P(none) = 0.65 \times 0.65 \times 0.65 \times 0.65$
$= 0.1785 = 17.85\%$

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Ex 22:

1. $P(A|B) = \frac{P(A)}{P(B)} = \frac{0.164}{0.25} = 0.656 = 65.6\%$

2. $P(\text{none}) = (1-0.25)\times(1-0.25) = 0.5625$

$P(\text{at least one}) = 1 - 0.5625 = 0.4375 = 43.75\%$